Modeling the Dynamics of Alzheimer's Disease

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What Alzheimer's Disease Is

Alzheimer's Disease (AD) is a neurodegenerative disease characterized by many complex interactions that affect over 50 million people worldwide. Many different aspects of the disease are anticipated, but it is not yet known how these factors interact [2].

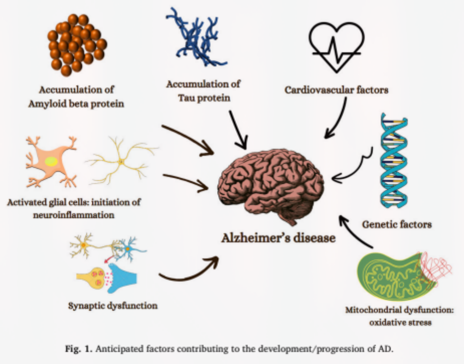

Figure: Various factors of AD from the literature

Hypotheses Galore

Various hypotheses have been proposed in order to understand the underlying mechanisms of AD; where the leading hypothesis for many years has been the *Amyloid Cascade Hypothesis*. However, there has been increasing support in the notion that it is insufficient in giving an effective description and treatment towards AD [1].

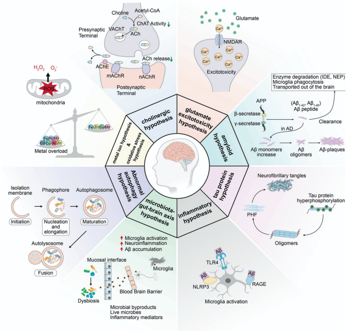

Figure: A visualization of the various hypotheses relating to the mechanisms of AD

Formation of a Newly Proposed Model

In order to understand these factors and subsequently the behavior of AD, many systems of ordinary differential equations (ODEs) have been proposed related to these hypotheses; but, a few are highlighted in regard to creating a newly proposed dynamical system:

- Observing the dynamics between Aβ and Ca^{2+} [3].
- Using causal models observing factors of genetic, non-amyloid dependent, and biomarker cascades [4].

The Proposed Model

Due to the feedback loop created by Aβ and Ca^{2+}, along with the interaction between the hyperphosphorylation of τ-protein caused by Aβ, a proposed causal model is described by the following system of ODEs:

$$\frac{dA_\beta}{dt} = a_1 + \underbrace{a_2 \frac{Ca}{Ca + \sigma}}_{\text{Holling Type II}} - r_1 A_\beta \tag{1}$$

$$\frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca \tag{2}$$

$$\frac{d\tau}{dt} = c_1 + \underbrace{c_2 A_\beta + c_3 Ca}_{\text{A}\beta\text{, Ca}^{2+}\text{ interaction}} - r_3 \tau \tag{3}$$

$$\frac{dN}{dt} = d_1 + d_2 \tau - k_4 N \tag{4}$$

$$\frac{dC}{dt} = e_1 + e_2 NR + e_3 \tau - k_5 C, \tag{5}$$

where $r_n = k_n + u_n$, and u_n is the treatment parameter.

On Invariance and Boundedness

The system can be shown to be *positively invariant*. This is shown by

$$\left.\frac{dA_\beta}{dt}\right]_{A_\beta=0}, \left.\frac{dCa}{dt}\right]_{Ca=0}, \left.\frac{d\tau}{dt}\right]_{\tau=0}, \left.\frac{dN}{dt}\right]_{N=0}, \left.\frac{dC}{dt}\right]_{C=0}.$$

Since the model follows biological scenarios, non-negative parameter choices mean that each compartment is non-negative, and thus the non-negative orthant is invariant

On Invariance and Boundedness

The system can also be shown to be *bounded* by

$$a_1 + a_2 \frac{Ca}{Ca + \sigma} - r_1 A_\beta \leq a_1 + a_2 - r_1 A_\beta < 0 \text{ if } A_\beta > \frac{a_1 + a_2}{r_1}$$
$$b_1 + b_2 A_\beta - r_2 Ca \leq b_1 + b_2 M - r_2 Ca < 0 \text{ if } Ca > \frac{b_1 + b_2 M}{r_2}.$$

since A_β, Ca feed into the other compartments it follows that each subsequent compartment is also bounded (Though due to time constraints this did not make it into the slides).

Discussion of Proposed Model

When analyzing the model, while equilibria are intractable, global asymptotic stability has been determined for the model. To show this, consider the first two compartments that are uncoupled from the rest of the proposed model:

$$\frac{dA_\beta}{dt} = a_1 + a_2\frac{Ca}{Ca+\sigma} - r_1A_\beta$$
$$\frac{dCa}{dt} = b_1 + b_2A_\beta - r_2Ca$$

Discussion of Proposed Model

For showing equilibrium, solve $dA_\beta/dt = 0$ in terms of A_β. Substituting this into dCa/dt and simplifying (using CAS), a quadratic is obtained in the form

$$-(r_1 r_2)Ca^2 + (a_1 b_2 + a_2 b_2 + b_1 r_1 - r_1 r_2 \sigma)Ca + a_1 b_2 \sigma + b_1 r_1 \sigma = 0.$$

Using *Descartes' Rule of Signs*, one positive root is determined, and hence a unique positive real equilibrium (A_β^*, Ca^*).

Discussion of Proposed Model

Then to show the non-existence of limit cycles, consider $\alpha(A_\beta, Ca) = 1$. It can be shown that

$$\operatorname{div} f = -r_1 - r_2 < 0 \implies \operatorname{div} \alpha f = -r_1 - r_2 < 0.$$

Since $\operatorname{div} \alpha f < 0$, by the Bendixson-Dulac Theorem there exists no limit cycles. This shows that the unique equilibrium for the uncoupled system is globally asymptotically stable.

Discussion of Proposed Model

Now, using (A_β^*, Ca^*), notice that

$$0 = c_1 + c_2 A_\beta + c_3 Ca - r_3 \tau$$
$$\tau = \frac{c_1 + c_2 A_\beta^* + c_3 Ca^*}{r_3}.$$

since a globally stable equilibrium has been determined for the uncoupled system, it follows that τ will also tend toward some equilibrium that is globally stable. This provides a similar cascade to N, C, and thus there exists a unique globally asymptotically stable equilibrium for the proposed model.

Potential Limitations

While global stability has been shown, there are a few drawbacks that have been noticed in regard to the model.

- The treatment terms do not seem to provide an accurate representation of the mechanisms of treatment/provide any new insights to literature.
- Fairly simplistic; would like to add more biologically descriptive interactions to model.

Future Directions

In order to provide a more interesting description of the mechanisms of AD, future directions include:

- Potential mass action terms added to the model.
- Sensitivity analysis of model.
- Optimal control for treatment terms.

References

[1] Kasper P Kepp et al. "The amyloid cascade hypothesis: an updated critical review". In: *Brain* 146.10 (2023), pp. 3969–3990.

[2] Mitali Maji and Subhas Khajanchi. "Mathematical models on Alzheimer's disease and its treatment: A review". In: *Physics of Life Reviews* 52 (2025), pp. 207–244.

[3] Swadesh Pal and Roderick Melnik. "Nonequilibrium landscape of amyloid-beta and calcium ions in application to Alzheimer's disease". In: *Physical Review E* 111.1 (2025), p. 014418.

[4] Jeffrey R Petrella et al. "Computational causal modeling of the dynamic biomarker cascade in Alzheimer's disease". In: *Computational and mathematical methods in medicine* 2019.1 (2019), p. 6216530.

Acknowledgments

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Closing

Thank you. Questions?